

GARVIT SINGH

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SUMMARY

PORTFOLIO: <https://portfolio-garvitsingh.vercel.app>

Full-stack developer (React, Node.js, Python) who builds and deploys AI-integrated web apps: a RAG assistant on Pinecone + Gemini and a Dockerized ML microservice. 500+ DSA problems solved.

PROJECTS

AgroInOne | React.js · Node.js · Python/Flask · SQL (Supabase) · Machine Learning · Random Forest

LIVE: <https://agroinone-main.onrender.com> GITHUB: <https://github.com/GarvitSid/AgroInOne>

- Built a two-stage ML advisory pipeline in Flask. A Random Forest classifier recommends a crop from soil (N, P, K, pH) and climate inputs (99.5% 5-fold CV accuracy, 2,200 samples, 22 crops). A Random Forest regressor then forecasts production for the chosen district and season.
- Resolved crop-name mismatches between the two datasets with a canonical mapping layer, and validated inputs (pH 0–14, non-negative nutrients, humidity 0–100%, area > 0) so bad data returns a clean 400 instead of a wrong prediction.
- Dockerized the ML service (Gunicorn with 4 threads, non-root user) and orchestrated React/Nginx, Express and Flask with Docker Compose and health checks. The Express gateway proxies to Flask over HTTP, and the regressor is capped at 30 trees and depth 12 to stay lightweight.
- Implemented JWT + bcrypt auth on Supabase with protected routes, and wrote 26 integration tests (auth, validation, ML proxy) wired into a GitHub Actions pipeline.

ChatMind AI College | React · Node.js/Express (TS) · MongoDB · Pinecone · Google Gemini API

LIVE: <https://chat-mind-ai.vercel.app> GITHUB: <https://github.com/GarvitSid/ChatMindAI>

- Built a full-stack RAG assistant. Admin-uploaded PDF/TXT files are parsed in memory, split into 1,000-character chunks (200 overlap), embedded (768-d, L2-normalized, batched with exponential-backoff retries) and indexed in Pinecone. Answers come from the top-3 chunks and show deduplicated source filenames.
- Added a 0.50 cosine-similarity gate so out-of-scope questions get a fixed refusal without calling the LLM, which saves API quota. In-scope answers use a context-only system prompt and a Gemini Flash → Flash-Lite fallback.
- Kept Pinecone and MongoDB consistent. Uploads index vectors first and roll them back on failure. Deletes remove vectors before the Mongo record so failures can be retried. Chat messages are saved only after generation succeeds, so outages leave no orphaned records.
- Secured the app with JWT, bcrypt, admin-only role middleware, and per-user session isolation (foreign and missing session IDs return identical 404s). It also has rate limits (5/min on auth, 15/min per user on chat) and a 5 MB PDF/TXT upload cap that rejects duplicate filenames.
- Wrote 43 Jest/Supertest tests on in-memory MongoDB with mocked Gemini and Pinecone, run in GitHub Actions. Built the admin dashboard and Zustand multi-session chat UI, deployed on Vercel and Render.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML5, CSS3, TypeScript; **Frontend:** React.js, Tailwind CSS, Bootstrap, Zustand, React Query, Vite; **Backend:** Node.js, Express.js, Flask, REST APIs, JWT Authentication; **Databases:** MongoDB, MySQL, PostgreSQL (Supabase), Pinecone (Vector DB); **Artificial Intelligence:** GenAI, AI Agents, LLMs, RAG, MCP, LangChain, Google Gemini API, scikit-learn; **DevOps & Tools:** Git, GitHub, Docker, Gunicorn, VS Code, Jest, Supertest, Github Action

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML)

Ajay Kumar Garg Engineering College (AKTU), Ghaziabad | Oct 2021 – Aug 2025

Relevant Coursework: Data Structures & Algorithms, Databases (SQL/MySQL), Software Engineering, Operating Systems, DBMS

VIRTUAL INTERNSHIPS & TRAINING

Frontend & AI Intern - CSRBOX / IBM SkillsBuild

Ghaziabad, UP (Remote) | July 2024 – Aug 2024

- Built a client-facing React dashboard and integrated REST APIs, improving responsiveness and reducing load times.
- Developed AI-assisted frontend features in a certified IBM SkillsBuild program covering applied AI fundamentals.

CCBP 4.0 (Continuous Career Building Program 4.0) Academy (Genius Plus)

NXTWave Disruptive Technologies Private Limited | July 2022 – Present

Relevant Coursework: JavaScript, React, Node.js, Express, SQL, Python, Git

CERTIFICATIONS

Model Context Protocol (MCP) Project – NxtWave (Aug 2025)

Skills: Prompt-driven AI workflows, Cursor IDE, Pipedream, MCP Servers.

AI Workflows & Automation – NxtWave (Jan 2026)

Skills: End-to-end automation, Make.com, AI agent integration.